

Khoshrav Piran Doctor

Computer Scientist & Robotics Researcher

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📍 Amherst, MA

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Summary

Ph.D. candidate in Computer Science at UMass Amherst specializing in robotics and autonomous decision making systems operating under uncertainty. Experienced in building real-world robotic systems, developing learning-based models, and deploying scalable software solutions.

Education

Ph.D. University of Massachusetts Amherst , Computer Science	Amherst, MA, USA
• Laboratory for Perceptual Robotics	2020 – 2026
M.S. University of Massachusetts Amherst , Computer Science	Amherst, MA, USA
• Laboratory for Perceptual Robotics	2017 – 2020
• Relevant Coursework: Robotics, Reinforcement Learning, Machine Learning, Artificial Intelligence, Algorithms for Data Science, Advance Algorithms, Feed-back Control Systems	
B.E. K. J. Somaiya College of Engineering , Information Technology	Mumbai, India
	2012 – 2016

Experience

Laboratory for Perceptual Robotics, UMass Amherst , Research Assistant	Amherst, MA, USA
• Developed learning-based exploration and model-learning methods for partially observable robotic systems, improving sample efficiency by up to 4× and enabling more effective belief space planning under uncertainty	Aug 2017 – present
• Investigated methods to learn reusable skills hierarchically to speed up grasping and transport of objects	8 years 9 months
• Implemented control systems for a 3 degree of freedom robotic head with an RGB-D camera	
• Performed routine diagnosis, maintenance and repair in the uBot robotic platform	
• Technologies used: ROS (C++ and Python development), Gazebo, Deep Learning, Reinforcement Learning	
Indiana University , Research Affiliate	Bloomington, IN, USA
• Analyzed IMU movement data to develop computational, non-invasive and objective approaches for detecting Autism Spectrum Disorder, Attention-Deficit/Hyperactivity Disorder and their comorbidity	Feb 2016 – present
• Designed Deep Learning and Machine Learning algorithms based on systems neuroscience to achieve 71% accuracy	10 years 3 months
• Published and presented relevant work at numerous physics and neuroscience meetings	
TRAC Labs, Inc. , Graduate Robotics Intern	Houston, TX, USA
• Developed multi-agent coordination systems to improve teleoperator situational awareness by leveraging additional heterogeneous robotic agents	Sept 2021 – Aug 2023
• Improved multi-agent path planning algorithms for real-time robotic systems	2 years
• Built visualization and debugging tools (RViz) to analyze planning behavior, system performance, and failure modes	
• Contributed towards experimental design and writing for research papers, reports, and documentation of projects	

AITOE Video Analytics Pvt. Ltd., Software Developer

- Designed and implemented online video analysis platform Newton for Search, Summarization and Statistics
- Developed and maintained aiSentinel Security Suite features like Camera Tampering Detection and Asset Tracking
- Researched technologies such as CMT to be implemented for Object Tracking
- Interacted directly with potential clients to identify and implement their requirements into our products
- Technologies used: C++, NodeJS, Python, PHP, Caffe, QT

Mumbai, India
July 2016 – July 2017
1 year 1 month

Computer Society of India, K. J. Somaiya College of Engineering Branch, Technical Head

- Developed content and trained council members to deliver workshops and seminars to the student body
- Responsible for management of the Technical Team of 4 individuals
- Responsible for the quality of the website and Android application of the Council
- Organized and delivered numerous workshops, seminars and technical competitions

Mumbai, India
July 2014 – June 2015
1 year 1 month

Publications

Deep learning diagnosis plus kinematic severity assessments of neurodivergent disorders

Doctor, Khoshrav P, McKeever, Chaundy, Wu, Di, Phadnis, Aditya, Plawecki, Martin H, Nurnberger Jr, John I, Jose, Jorge V
www.nature.com/articles/s41598-025-04294-9 (Nature Scientific Reports)

Deep Learning and Statistical longitudinal medication studies of attention-deficit/hyperactivity disorder of (ADHD) participants

Jose, Jorge V, Doctor, Khoshrav, McKeever, Chaundy Lee
schedule.aps.org/smt/2025/events/MAR-G65/1 (APS March Meeting Abstracts)

A family of neurodevelopmental biomarkers extracted from a statistical analysis of kinematic data measured with high-definition sensors

Jose, Jorge, Doctor, Khoshrav, Phadnis, Aditya, McKeever, Chaundy, Plawecki, Martin
ui.adsabs.harvard.edu/abs/2023APS..MARA11013J/abstract (APS March Meeting Abstracts)

Deep Learning Diagnostic Analysis of the kinematic movement data recorded from individuals with Neurodevelopmental Disorders

Doctor, Khoshrav, Wu, Di, Phadnis, Aditya, Plawecki, Martin, Nurnberger Jr, John, Jose, Jorge
ui.adsabs.harvard.edu/abs/2022APS..MARD03003D/abstract (APS March Meeting Abstracts)

Learning Intrinsically Motivated Transition Models for Autonomous Systems

Doctor, Khoshrav, Ghosh, Hia, Grupen, Roderic
doi.org/10.1109/ICDL53763.2022.9962188 (2022 IEEE International Conference on Development and Learning (ICDL))

Evaluating Sensorimotor Abstraction on Curricula for Learning Mobile Manipulation Skills

Youngquist, Oscar, Spiro, Alenna, Doctor, Khoshrav, Grupen, Roderic
doi.org/10.1109/ICDL53763.2022.9962221 (2022 IEEE International Conference on Development and Learning (ICDL))

Statistical Motor Biomarkers Characterizing age-dependence in Neurodevelopmental Disorders

Doctor, Khoshrav, Wu, Di, Phadnis, Aditya, Nurnberger, John, Plawecki, Martin, Jose, Jorge
ui.adsabs.harvard.edu/abs/2021APS..MARS12015D/abstract (APS March Meeting Abstracts)

Demystifying multi-faceted video summarization: tradeoff between diversity, representation, coverage and importance

Kaushal, Vishal, Iyer, Rishabh, Doctor, Khoshrav, Sahoo, Anurag, Dubal, Pratik, Kothawade, Suraj, Mahadev, Rohan, Dargan, Kunal, Ramakrishnan, Ganesh
arxiv.org/abs/1901.01153 (2019 IEEE Winter Conference on Applications of Computer Vision (WACV))

Learning From Less Data: A Unified Data Subset Selection and Active Learning Framework for Computer Vision

Kaushal, Vishal, Iyer, Rishabh, Kothawade, Suraj, Mahadev, Rohan, Doctor, Khoshrav, Ramakrishnan, Ganesh
doi.org/10.1109/WACV.2019.00142 (2019 IEEE Winter Conference on Applications of Computer Vision (WACV))

Learning from less data: Diversified subset selection and active learning in image classification tasks

Kaushal, Vishal, Sahoo, Anurag, Doctor, Khoshrav, Raju, Narasimha, Shetty, Suyash, Singh, Pankaj, Iyer, Rishabh, Ramakrishnan, Ganesh
arxiv.org/abs/1805.11191 (arXiv preprint arXiv:1805.11191)

A unified multi-faceted video summarization system

Sahoo, Anurag, Kaushal, Vishal, Doctor, Khoshrav, Shetty, Suyash, Iyer, Rishabh, Ramakrishnan, Ganesh
arxiv.org/abs/1704.01466 (arXiv preprint arXiv:1704.01466)

PowerShare: A distributed processing system for Android devices

Kumar, Vignesh Pradeep, Jayawant, Shubham, Doctor, Khoshrav, Vidhani, Sanjay
doi.org/10.1109/CESYS.2016.7889989 (2016 International Conference on Communication and Electronics Systems (ICCES))

Skills

Programming: C, C++, Python, Java, MATLAB, Arduino, Processing

Robotics & ML: ROS, Reinforcement Learning, Deep Learning, Planning, SolidWorks

Web Development: HTML, JavaScript, CSS, PHP

Databases: SQL, MongoDB

Interests

Robotics: Autonomous Systems, Active Exploration, Robot Learning, POMDPs, Multi-Agent Systems

Computational Neuroscience: Neurodevelopmental Disorders, Autism Spectrum Disorder, Attention-Deficit/Hyperactivity Disorder

Projects

Spatial Model Building Using a Robotic Head

July 2017 – Dec 2017

- Built probabilistic spatial models using deep learning
- Used models for goal-directed action selection

PowerShare: A Distributed Processing System for Android Devices

Jan 2015 – July 2016

- Developed distributed processing system for Android devices using Bluetooth and WiFi-Direct
- Implemented task scheduling and load balancing algorithms
- Evaluated system performance on real-world applications

Teaching

Teaching Assistant, University of Massachusetts Amherst	Aug 2017 – Dec 2025
• Platform for Ethical and Responsible Computing (Fall 2024, Spring 2025, Fall 2025) • COMPSCI 589 (Spring 2024): Machine Learning • COMPSCI 683 (Spring 2023): Artificial Intelligence • COMPSCI 187 (Fall 2022): Programming with Data Structures • COMPSCI 603 (Spring 2020): Robotics • COMPSCI 503 (Fall 2018): Embedded Systems Engineering • COMPSCI 240 (Spring 2018): Reasoning Under Uncertainty • COMPSCI 220 (Fall 2017): Programming Methodology	

Awards

Summer 2024 Dissertation Writing Fellowship	June 2024
Fall 2023 Proposal Writing Fellowship	Aug 2023
1st Place Project Competition	Mar 2016
2nd Place Project Competition	Mar 2016
Highest Committed Student Branch Activist	Dec 2014
Jewel of Somaiya Award for Outstanding Performance	Dec 2013